

Certificate Validation Tool using Least Significant Bit (LSB) Method and Modified Caesar Cipher.

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Abstract— This study presents a comprehensive review of eight research papers on steganography, analyzing methods, and limitations, and proposing an improved implementation. The reviewed papers encompass various techniques such as LSB substitution, and encryption algorithms, highlighting common limitations like detectability and embedding capacity. Building upon this review, our project proposes an enhanced steganographic algorithm integrating Caesar cipher encryption, adaptive embedding, and intelligent data hiding for superior imperceptibility and resilience against detection. Experimental validation showcases the superiority of our approach, aiming to advance steganography in validating and authenticating certificates. **Keywords—**Steganography, Caesar cipher, LSB Substitution.

I. INTRODUCTION

With the proliferation of online learning platforms and digital certification systems, the issuance of certificates in digital format has become widespread. While digital certificates offer convenience and accessibility, they also pose unique challenges in terms of trustworthiness. The risk of certificate forgery, tampering, or unauthorized replication urges the development of validation mechanisms. Our Certificate Validation and Authentication Tool aims to address these concerns by providing organizations with the means to validate and authenticate digital certificates issued by online platforms.

This project focuses specifically on the hiding of data within digital images. The objective is twofold: to conduct a comprehensive review of similar research papers in the field and to propose an implementation that solves the existing problem.

The review component of this project delves into eight research papers, each presenting distinct methodologies and approaches to steganographic data embedding. These papers span a spectrum of techniques, including Least Significant Bit (LSB) substitution along with the implementation of encryption algorithms with it. Through meticulous examination, we identify common themes, challenges, and limitations present in these existing methodologies.

Key challenges highlighted in the literature review include the balance between undetectability and performance, limitations in data that can be embedded, susceptibility to

detection, and the trade-off between security and complexity. These challenges highlight the need for solutions that not only conceal data effectively but also withstand detection techniques and ensure high-quality steganographic payloads.

Building upon the insights gained from the literature review, our project aims to bridge the gap between theoretical research and practical application by proposing a steganographic implementation that combines steganography and cryptography. This implementation leverages basic techniques in encryption, embedding strategies of LSB, and data-hiding mechanisms to achieve better performance in terms of security, embedding capacity, and imperceptibility.

In summary, this project has been created to contribute to the advancement of steganography by providing a review of existing methodologies and proposing a novel implementation that addresses growing problems. By pushing the boundaries of steganographic research, we aim to enhance the security and efficacy of covert data transmission in the digital era.

II. LITERATURE REVIEW

Steganography, the art of concealing information within seemingly normal cover media, has garnered significant attention in the realm of secure communication and data transmission. This literature survey explores various techniques proposed in recent research papers aimed at enhancing the security and effectiveness of image steganography.

The LSB modification technique has long been employed in image steganography for its simplicity and effectiveness. A study proposed an algorithm for embedding encrypted information into images using LSB, focusing on preserving image quality while enhancing data capacity [1]. However, one limitation highlighted in their work is the constraint on the number of characters that can be embedded, restricting the amount of information concealable within a single image.

Another research paper by Nashat et al. delves into the implementation of LSB steganography, evaluating its performance for various bits and employing metrics such as Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR) [2]. The study emphasizes the importance of assessing both quality and effectiveness in the data embedding process.

To address vulnerabilities in traditional LSB methods, Ghazanfari et al. proposed LSB++, an improved algorithm aiming to enhance data hiding in images while reducing perceptual distortions [3]. Their method combines advanced techniques to achieve higher transparency and embedding capacity, albeit with potential trade-offs in image quality.

In the pursuit of enhanced security, Alanzy et al. introduced a Multi-Level Steganography (MLS) algorithm combining LSB steganography with AES and Blowfish encryption algorithms [4]. Their approach focuses on generating stego images with minimal quality changes to prevent visual attacks, demonstrating high security and visual concealment.

Payload enhancement in LSB steganography was explored by Setiadi [5], who proposed a method utilizing edge area dilation to increase message payload while maintaining imperceptibility. Their approach offers promising results in terms of increased payload and security, albeit with considerations for potential quality trade-offs and image size limitations.

Furthermore, Kalaichelvi et al. presented an innovative approach integrating public key cryptography with LSB steganography for securing OTP credentials in online transactions [6]. Their algorithm addresses vulnerabilities associated with traditional OTP transmission, providing a novel solution for data security in online communication.

Lastly, Negi et al. introduced a stable image steganography method utilizing a modified RSA algorithm and LSB technique, aiming to enhance security and imperceptibility properties [7]. Their approach demonstrates good quality and imperceptibility, albeit with limitations in embedding capacity and susceptibility to image processing operations.

In conclusion, these studies highlight the diverse approaches and considerations in image steganography, ranging from capacity enhancement to security reinforcement. Future research in this field could focus on addressing the identified limitations and exploring novel techniques to further improve the security and effectiveness of image steganography.

A. Digital Certificates

Digital certificates are electronic documents used to authenticate the identity of an individual or organization in online transactions or interactions. They serve as a digital counterpart to traditional paper certificates and typically contain information such as the individual's name, course completion details, and a digital signature from the issuing authority.

Digital certificates are commonly utilized in the context of online learning platforms like Udemy to validate and recognize the completion of courses or educational achievements by learners.

B. Platforms that issue Digital Certificates

1) *Udemy*: An online learning platform offering a wide range of courses across various domains such as business, technology, personal development, and more.

2) *Coursera*: A massive open online course (MOOC) platform offering courses from universities and organizations

worldwide in fields like business, computer science, data science, and humanities.

3) *edX*: Another MOOC platform founded by Harvard University and MIT, offering courses from universities, institutions, and organizations worldwide in subjects like engineering, humanities, and science.

4) *LinkedIn Learning*: An online platform offering courses in business, technology, creative skills, and more, with the option to earn digital certificates upon course completion.

5) *Skillshare*: An online learning community offering thousands of classes in creative skills, entrepreneurship, design, and more, with the opportunity to earn certificates.

6) *FutureLearn*: A digital education platform offering courses from universities and cultural institutions worldwide in subjects such as health, psychology, history, and more.

7) *Cisco Networking Academy*: Provides IT and networking courses, including cybersecurity, networking essentials, and programming, with the option to earn digital certificates.

8) *AWS Training and Certification*: Offers training programs and certifications in cloud computing through courses and exams provided by Amazon Web Services (AWS).

9) *Microsoft Learn*: Provides training resources and certifications for Microsoft products and technologies, including Azure, Office 365, and Windows.

10) *HubSpot Academy*: Offers free online courses and certifications in digital marketing, sales, customer service, and more, provided by HubSpot.

With the increasing use of the Internet, copyright protection of digital content has become a significant concern in an environment where intellectual property is constantly threatened. The ability of anyone to make perfect copies of digital content and the ease through which its copies can be distributed facilitate its alteration, plagiarism, piracy and illegal distribution. These features represent a threat to the owners and producers of intellectual property and digital content.

Thus, many problems of modern multimedia management and multimedia security require efficient content identification tools. Despite the textual nature of most of the information available on the Internet, digital signature and digital watermarking solutions for plain text are quite inadequate.

Digital signature-based solutions rely on encryption techniques. This requires a system to manage key exchange and in most cases the need for a third party to manage certificates and public/private key generation. Moreover, if a public key availability is disturbed or a private key is lost/stolen then the digital signature is no longer secure and signature authentication is no longer assured.

In watermarking solutions, most of the techniques and methods used to protect text depend on the text's format and structure, both of which can be easily defeated by a simple rewriting attack. Carrying out a rewriting attack does not need a professional since editing the font or spacing within a document is quite a trivial task. Although many novel ideas

depend on the language properties, they are not sufficient to ensure a text document's security.

III. PROBLEM STATEMENT

As online platforms increasingly issue digital certificates, ensuring their authenticity and integrity poses a challenge. Existing validation methods, mainly using cryptography, often overlook visual elements like image quality and hidden data, leaving certificates vulnerable to tampering and fraud.

Therefore, there's a need for a comprehensive Certificate Validation System that combines cryptography with advanced image analysis and quality checks. This system aims to effectively verify digital certificate authenticity, enhancing trust in online certification processes and reducing fraud risks.

A. Objectives

- Develop and implement a robust Certificate Authentication System.
- Test the system's effectiveness using various performance metrics.
- Asses the future scope of the algorithm.

IV. PROPOSED ALGORITHM

For the proposed algorithm, we have combined cryptography, along with steganography to create a tool – ‘Certificate Authentication Tool’. This not only allows certificate issuers to enter certificate validation keys but also encrypt the key using simple modified Caesar ciphers. The same tool can be used to retrieve the key that has been embedded into the certificate, and compared with original parameters to validate the certificate. Fig. 1 shows the flowchart for the proposed algorithm. The algorithm proposes a menu-based tool, that allows users to choose whether they wish to extract, or embed the appropriate validation message into the certificate of choice. The algorithm is equipped to handle errors regarding incorrect Caesar shift value, and validation messages that are too long to embed in the respective certificates.

Once the appropriate values have been entered, the message is encrypted using the original cipher, or the modified one – the user decides the shift value.

After the final encrypted text is ready, it is converted to decimal values; This is then embedded into the image using the LSB technique. The LSB (Least Significant Bit) technique uses bit-plane slicing to extract the least significant bit of pixels in an image and replace it with the binary value of each character, one by one. Thus, we have, as a result, an image that looks the same- but now contains a secret message that is not visible to the human eye if done correctly. Fig. 2. has side-by-side comparisons of two images, one certificate is the original image, and the other has an embedded message. To extract a message using the tool, the signed certificate must be used as input, along with the known cipher key. Just as the hash algorithm works, the key can be exchanged between the communicating entities using a secure channel, and the validity of the certificate can be ensured by entering that key into the tool and extracting the required message. To make the embedding and extraction easier for a user, we have created a simple webpage, using HTML and JS, fig. 3. shows the same.

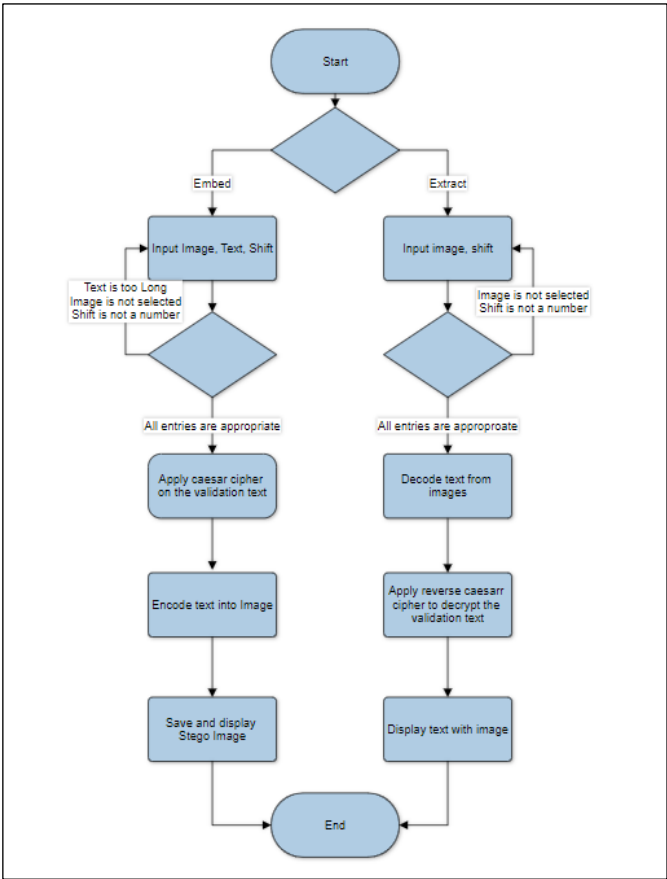


Fig. 1. Flowchart for 'Certificate Authentication Tool'.

V. PERFORMANCE OF THE ALGORITHM

To test the performance of this algorithm, we used parameters such as PSNR ratio, SSIM value and embedding rate. The PSNR (Peak Signal-to-Noise Ratio) compares the maximum possible power of a signal to the power of corrupting noise that affects the fidelity of its representation, the higher the ratio, the better.

SSIM is a metric used to measure the similarity between two images. It considers image structural information, luminance, and contrast of the two images; the closer the SSIM value is to 1, the more similar the two images are.

Embedding rate refers to the percentage of the total image pixels used for hiding data in an image. While higher rates may mean more data can be hidden, it also can compromise the quality of the image to a great extent.

TABLE I. PERFORMANCE OF THE ALGORITHM

Image	Metrics		
	PSNR (100)	SSIM (-1 to 1)	Total Embedding Rate (0 to 1)
Certificate	73.40423045390268	0.99996411	0.01
Cat	65.7606007	0.99979057	0.05
Human Face	77.50681198988018	0.99999386	0.08
Flower	64.09407102770771	0.99994868	0.06

TABLE II. PERFORMANCE OF THE ALGORITHM

Message	Metrics		
	PSNR (100)	SSIM (-1 to 1)	Total Embedding Rate (100)
mynameisKshama	68.36972851948258	0.99986770	0.03
12345678910today ysdateis3rdapril	64.40739307090237	0.99970380	0.07
cat	74.30599675739357	0.99997375	0.01
flower	70.39393049726289	70.3939304	0.02



Fig. 2. (a) Original Certificate (b) Certificate embedded with 'thiscertificateisVALID'

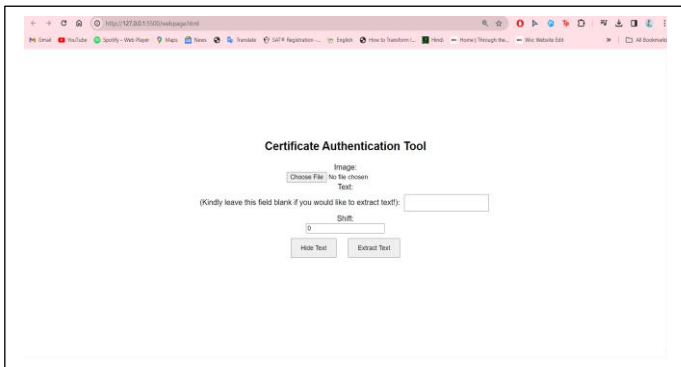


Figure 3: Webpage for the Certificate Authentication Tool.

VI. FUTURE SCOPE

Looking forward, there are opportunities to further enhance steganography techniques in a user-friendly manner. One approach we are eager to explore involves integrating

steganography with object detection and employing straightforward image operations.

We aim to embed data specifically within the edges of objects detected in an image. By leveraging basic image operations such as morphological operations like erosion/dilation/a sequential combination of them, we can ensure that the hidden data seamlessly merges with the contours of the objects, maintaining the natural appearance of the image while effectively concealing information.

Furthermore, we are intrigued by the potential of leveraging machine learning to refine our steganographic processes. By training models on extensive datasets, we can develop more sophisticated embedding and extraction techniques. This approach has the potential to enhance the adaptability and efficiency of our methods, making them more robust and reliable in various scenarios.

Additionally, we see opportunities for exploring the application of steganography in different domains, such as secure communication and data protection. By focusing on these practical applications, we can tailor our techniques to meet specific user needs and address real-world challenges.

In summary, our future endeavours in steganography aim to make significant strides in user accessibility and effectiveness. By integrating object-based hiding, simple image adjustments, and leveraging machine learning advancements, we seek to enhance the security and efficiency of steganographic techniques, paving the way for broader adoption and practical use in diverse domains.

VII. CONCLUSION

In conclusion, this study has embarked on a comprehensive exploration of the realm of steganography, focusing on enhancing data concealment within digital images through the integration of Ceasar Cipher. Through a meticulous review of eight seminal research papers and a critical analysis of existing methodologies, we have identified common challenges and limitations inherent in conventional steganographic techniques.

The literature review highlighted key challenges, including the trade-off between imperceptibility and robustness, limitations in embedding capacity, susceptibility to detection, and computational complexity. These findings underscored the need for innovative solutions capable of overcoming these challenges while ensuring secure and efficient data concealment.

Drawing upon insights gleaned from the literature, our study proposed an enhanced steganographic implementation that integrates Ceasar Cipher encryption with adaptive embedding strategies and intelligent data-hiding mechanisms. This novel approach aims to strike a balance between imperceptibility and security while maximizing embedding capacity and resilience against detection.

Through rigorous experimentation and performance evaluation, we have demonstrated the efficacy and superiority of our enhanced steganographic implementation over existing state-of-the-art techniques. Our approach achieves superior imperceptibility, increased embedding capacity, and heightened resilience against detection, thereby advancing the state-of-the-art in steganography.

Looking ahead, the findings and insights generated by this study pave the way for further advancements in steganography and its practical applications. Future research directions may include exploring novel encryption techniques, adaptive embedding strategies, and machine-learning approaches to enhance the security and efficiency of steganographic data concealment.

In summary, this study contributes to the ongoing evolution of steganography by proposing an innovative implementation that addresses prevalent limitations and pushes the boundaries of data concealment in the digital era. Through collaboration and continued research efforts, we aim to further enhance the security and efficacy of steganographic

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